

A Thin-Cross Counterexample to a Star-Body Product-Section Inequality

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Status: scoped candidate counterexample, likely valid. The “or even star” strengthening in Question 12 is false. The construction is nonconvex and does not settle the convex-body or L_p , $p > 2$, parts of the question.

1 Source question

Eskenazis asks the following on PDF page 13 of the source paper [1]. The paper is *On extremal sections of subspaces of L_p* , arXiv:1806.04333.

It is conceivable that the reverse inequality of (86) holds true when $p > 2$.

Question 12. Let $p \in (2, \infty]$ and $X = (\mathbb{R}^m, \|\cdot\|_X)$ which is isometric to a subspace of L_p . Is it true that

$$|B_p^{n-1}(X)| \leq |B_p^n(X) \cap H_\theta| \quad (87)$$

for every unit vector θ in $\mathbb{R}^n$? In particular, does every symmetric convex (or even star) body $K \subseteq \mathbb{R}^m$ satisfy

$$|K|^{n-1} \leq |K^n \cap H_\theta| \quad (88)$$

for every unit vector θ in $\mathbb{R}^n$?

Figure 1: Question 12 as rendered from page 13 of the source paper.

The scoped clause considered here asks whether every symmetric star body $K \subseteq \mathbb{R}^m$ satisfies

$$|K|^{n-1} \leq |K^n \cap H_\theta| \quad (1)$$

for every unit $\theta \in \mathbb{R}^n$, where

$$H_\theta = \{(x_1, \dots, x_n) \in (\mathbb{R}^m)^n : \theta_1 x_1 + \dots + \theta_n x_n = 0\}.$$

2 Proof intuition

A very thin cross is a mixture of two perpendicular arms. In the diagonal section of K^3 , three points of the cross must sum to zero. If all three occupy the same arm, this is possible with order- δ^2 volume. Mixed-arm configurations are smaller, of order δ^3 . The ambient normalization $|K|^2$ is also of order δ^2 , but the limiting ratio is only $27/32 < 1$.

The proof makes this heuristic exact. Inclusion-exclusion writes the cross indicator as the sum of its two rectangular-arm indicators minus their square overlap. Every resulting four-dimensional integral factors into two one-dimensional interval-convolution areas, giving an explicit polynomial in the arm thickness.

3 The counterexample

Fix

$$\delta = \frac{1}{100}, \quad H = [-1, 1] \times [-\delta, \delta], \quad V = [-\delta, \delta] \times [-1, 1],$$

and put $K = H \cup V \subseteq \mathbb{R}^2$. Let $n = 3$ and $\theta = (1, 1, 1)/\sqrt{3}$, so

$$H_{\text{diag}} = \{(x_1, x_2, x_3) \in (\mathbb{R}^2)^3 : x_1 + x_2 + x_3 = 0\}.$$

Theorem 1. *The set K is a centrally symmetric star body, and*

$$\frac{|K^3 \cap H_{\text{diag}}|}{|K|^2} = \frac{547083}{633616} < 1.$$

Consequently, inequality (1) is false for symmetric star bodies.

Proof. Step 1: geometric preliminaries. Both H and V are compact, centrally symmetric, convex, and contain the origin. Their union is therefore compact, centrally symmetric, and star-shaped about the origin. It contains the square $[-\delta, \delta]^2$ in its interior, and its radial function is the maximum of the continuous radial functions of H and V . Thus K is a star body. Writing $S = H \cap V = [-\delta, \delta]^2$, inclusion-exclusion gives

$$|K| = |H| + |V| - |S| = 8\delta - 4\delta^2. \quad (2)$$

The map

$$T : (\mathbb{R}^2)^2 \longrightarrow H_{\text{diag}}, \quad T(x, y) = (x, y, -x - y),$$

has Gram matrix $\begin{pmatrix} 2 & 1 \\ 1 & 2 \end{pmatrix}$ in each coordinate. Its four-dimensional Jacobian is therefore $\sqrt{3^2} = 3$. Hence

$$|K^3 \cap H_{\text{diag}}| = 3I_\delta, \text{quad} I_\delta = \int_{\mathbb{R}^2} \int_{\mathbb{R}^2} \mathbf{1}_K(x) \mathbf{1}_K(y) \mathbf{1}_K(x+y) dx dy. \quad (3)$$

Step 2: the interval calculation. For positive a, b, c , define

$$J(a, b, c) = \int_{|u| \leq a} \int_{|v| \leq b} \mathbf{1}_{\{|u+v| \leq c\}} du dv.$$

If $r \geq s \geq t$ is the decreasing rearrangement of a, b, c , slicing the convolution of two centered intervals gives

$$J(a, b, c) = \begin{cases} 4st, & r \geq s + t, \\ 2(rs + rt + st) - (r^2 + s^2 + t^2), & r < s + t. \end{cases} \quad (4)$$

Indeed, the first case is the plateau regime of the convolution trapezoid; the second subtracts its two corner triangles.

Now

$$\mathbf{1}_K = \mathbf{1}_H + \mathbf{1}_V - \mathbf{1}_S.$$

Expanding the three factors in (3) gives 27 terms. A term with rectangles $R(a_1, b_1)$, $R(a_2, b_2)$, $R(a_3, b_3)$ factors as

$$J(a_1, a_2, a_3)J(b_1, b_2, b_3).$$

Using (4), and grouping permutations of H, V, S , gives the complete sum below; the signs include the copies of $-\mathbf{1}_S$.

Grouped multisets	total contribution to I_δ
HHH and VVV	$18\delta^2$
HHV and HVV	$96\delta^3 - 24\delta^4$
HHS and VVS	$-72\delta^3 + 18\delta^4$
HVS	$-96\delta^4$
HSS and VSS	$72\delta^4$
SSS	$-9\delta^4$
Total	$18\delta^2 + 24\delta^3 - 39\delta^4$

Thus, by (2) and (3),

$$\frac{|K^3 \cap H_{\text{diag}}|}{|K|^2} = \frac{3(18 + 24\delta - 39\delta^2)}{16(2 - \delta)^2}. \quad (5)$$

At $\delta = 1/100$, exact rational arithmetic yields

$$I_\delta = \frac{182361}{100000000}, \quad |K| = \frac{199}{2500}, \quad \frac{3I_\delta}{|K|^2} = \frac{547083}{633616} < 1.$$

This contradicts (1). □

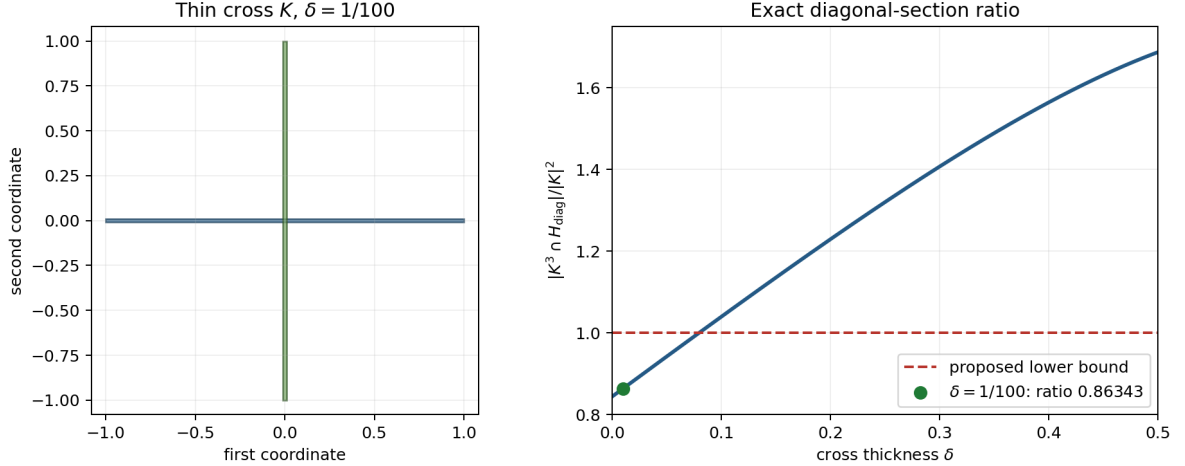


Figure 2: The thin-cross construction and its exact ratio from (5).

4 Verification

The script `code/verify_star_counterexample.py` evaluates all 27 terms using rational arithmetic, confirms the polynomial for I_δ and the final fraction exactly, and generates the figure. It reports `closed_form_match=True` and PASS. Commands and output are recorded in `verification.md`; computation is not used as a substitute for the proof.

5 Scope and bounded novelty check

This theorem answers only the explicit “or even star” strengthening in Question 12. The cross is not convex. No claim is made about symmetric convex bodies or unit balls of subspaces of L_p for $p > 2$.

The run indexes were searched by arXiv id, title, question number, and core terms. A bounded primary-source web/arXiv search used the exact question language, “star body,” “product section,” and “block hyperplane.” It found the source and general section-volume work but no later explicit answer or matching thin-cross calculation. This is not an exhaustive novelty claim.

An exploratory exact-polytope search tested standard and random centrally symmetric convex bodies in dimensions two and three and found no violation. That finite search has no bearing on whether the convex case is true.

6 Human-review recommendation

Independent review should check the Jacobian factor 3, formula (4), and the grouped inclusion-exclusion table. The result should remain explicitly scoped to star bodies unless a separate convex example or proof is found.

References

- [1] A. Eskenazis, *On extremal sections of subspaces of L_p* , Discrete Comput. Geom. **65** (2021), 489–509; arXiv:1806.04333.